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Q1: Addition of numbers

Welcome addition.py

addition.py

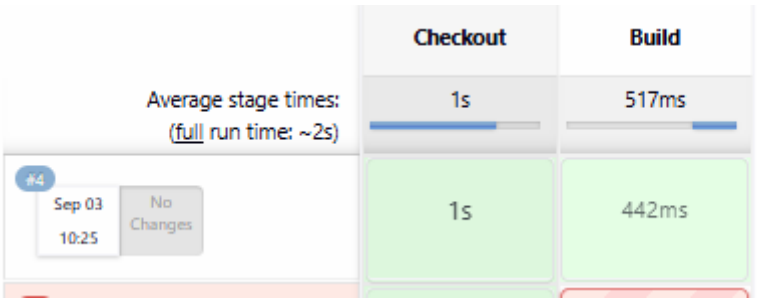
```python
import sys
def add_numbers(a, b):
    return a + b
if __name__ == "__main__":
    num1 = int(sys.argv[1])
    num2 = int(sys.argv[2])
    result = add_numbers(num1, num2)
    print("================================")
    print("Addition Result")
    print("================================")
    print(f"First Number : {num1}")
    print(f"Second Number: {num2}")
    print(f"Sum          : {result}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\batch1\Desktop\practice> git init
Initialized empty Git repository in C:/Users/batch1/Desktop/practice/.git/
PS C:\Users\batch1\Desktop\practice> git add .
PS C:\Users\batch1\Desktop\practice> git commit -m 'add the two numbers'
[master (root-commit) 483c6a6] add the two numbers
 1 file changed, 13 insertions(+)
 create mode 100644 addition.py
PS C:\Users\batch1\Desktop\practice> git remote add origin https://github.com/Nirmal-Priya/
PS C:\Users\batch1\Desktop\practice> git remote add origin https://github.com/Nirmal-Priya/
PS C:\Users\batch1\Desktop\practice> git branch -m main
PS C:\Users\batch1\Desktop\practice> git branch -M  main
PS C:\Users\batch1\Desktop\practice> git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 20 threads
```

```
 > git.exe config remote.origin.url https://github.com/Nirmal-Priya/practice.git # timeout=10
Fetching upstream changes from https://github.com/Nirmal-Priya/practice.git
 > git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/Nirmal-Priya/practice.git +refs/heads/*:refs/remotes/origin/* # timeout=10
Seen branch in repository origin/main
Seen 1 remote branch
 > git.exe show-ref --tags -d # timeout=10
Checking out Revision 483c6a65e884648997f05b38adc6aaf3ea49d561 (origin/main)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 483c6a65e884648997f05b38adc6aaf3ea49d561 # timeout=10
Commit message: "add the two numbers"
 > git.exe rev-list --no-walk 483c6a65e884648997f05b38adc6aaf3ea49d561 # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\practice>python addition.py 10 20
================================
Addition Result
================================
First Number : 10
Second Number: 20
Sum          : 30
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```



```
pipeline{
    agent any
     parameters {
        string(name: 'BRANCH_NAME',
               defaultValue: 'main',
               description: 'Git branch')
        string(name: 'NUM1',
               defaultValue: '10',
               description: 'First Number')
        string(name: 'NUM2',
               defaultValue: '20',
               description: 'Second Number')
    }
    stages{
        stage('Checkout'){
            steps{
                checkout scmGit(branches: [[name: '**']], extensions: [],
userRemoteConfigs: [[url: 'https://github.com/Nirmal-Priya/practice.git']])
            }
        }
        stage('Build'){
            steps{
                bat "python addition.py ${params.NUM1} ${params.NUM2}"
            }
        }
    }
}
```

```
pipeline{
    agent any
     parameters {
        string(name: 'BRANCH_NAME',
               defaultValue: 'main',
               description: 'Git branch')
        string(name: 'NUM1',
               defaultValue: '10',
               description: 'First Number')
        string(name: 'NUM2',
               defaultValue: '20',
               description: 'Second Number')
    }
    stages{
        stage('Checkout'){
```

Use Groovy Sandbox ?

ipeline Syntax

Q2: Calculator of addition, subtraction, multiplication, division

Welcome | calculator.py

calculator.py

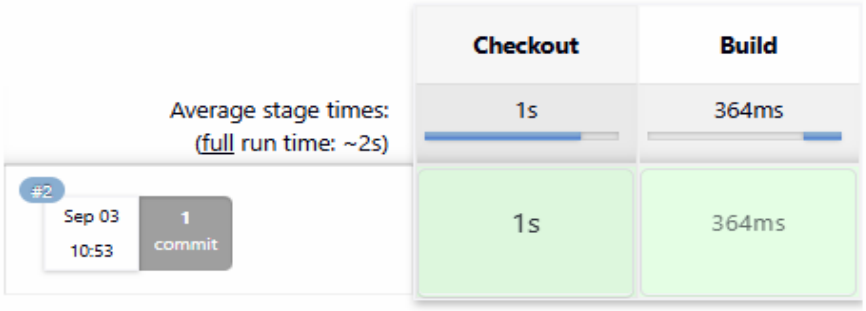
```
import sys
a=float(sys.argv[1])
b=float(sys.argv[2])
print("Addition=",a+b)
print("Subraction=",a-b)
print("Multiplication=",a*b)
print("Division=",a/b)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\batch1\Downloads\exercise2> git commit -m 'example calculators'
[main 63d4d67] example calculators
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\batch1\Downloads\exercise2> git branch -M main
PS C:\Users\batch1\Downloads\exercise2> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 296 bytes | 296.00 KiB/s, done
```

exercise2

Stage View

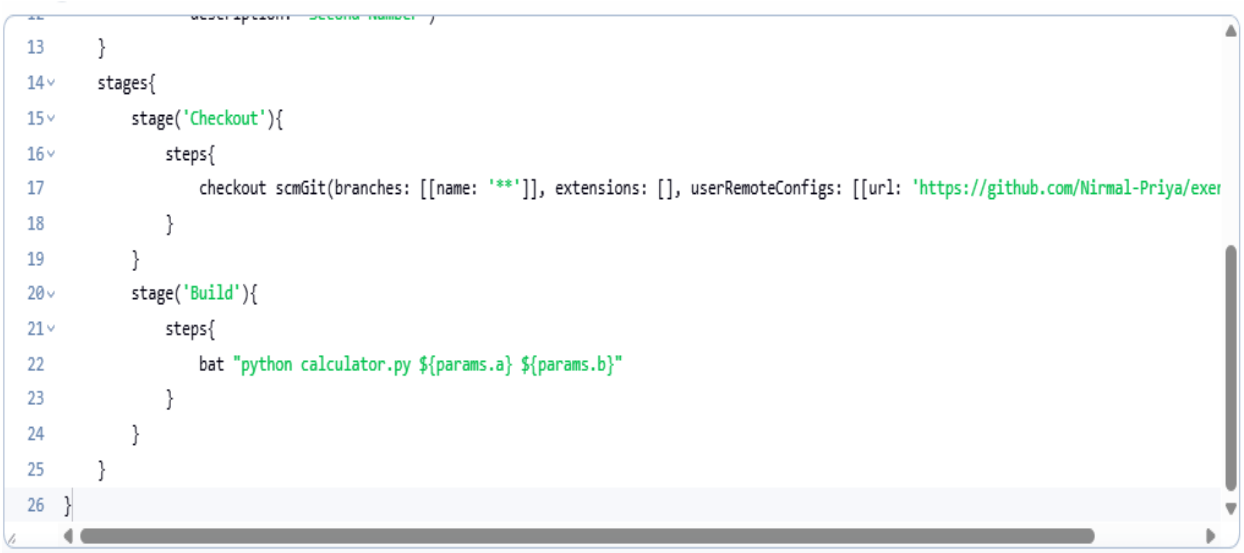


Permalinks

- Last build (#2), 2 min 9 sec ago
- Last stable build (#2), 2 min 9 sec ago
- Last successful build (#2), 2 min 9 sec ago
- Last completed build (#2), 2 min 9 sec ago

```
 > git.exe show-ref --tags -d # timeout=10
Checking out Revision 63d4d670d94778a0d7f82f565e47f2c7cc342a8c (origin/main)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 63d4d670d94778a0d7f82f565e47f2c7cc342a8c # timeout=10
Commit message: "example calculators"
 > git.exe rev-list --no-walk 56444f823dfff8834ea46a27324f614f876d488d # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\exercise2>python calculator.py 10 20
Addition= 30.0
Subraction= -10.0
Multiplication= 200.0
Division= 0.5
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```



```
    }
    stages{
        stage('Checkout'){
            steps{
                checkout scmGit(branches: [[name: '**']], extensions: [], userRemoteConfigs: [[url: 'https://github.com/Nirmal-Priya/exer
            }
        }
        stage('Build'){
            steps{
                bat "python calculator.py ${params.a} ${params.b}"
            }
        }
    }
}
```

```
pipeline{
   agent any
    parameters {
      string(name: 'BRANCH_NAME',
            defaultValue: 'main',
            description: 'Git branch')
      string(name: 'a',
            defaultValue: '10',
            description: 'First Number')
      string(name: 'b',
            defaultValue: '20',
            description: 'Second Number')
   }
   stages{
      stage('Checkout'){
         steps{
            checkout scmGit(branches: [[name: '**']], extensions: [],
userRemoteConfigs: [[url: 'https://github.com/Nirmal-Priya/exercise2.git']])
         }
      }
      stage('Build'){
         steps{
            bat "python calculator.py ${params.a} ${params.b}"
         }
      }
   }
}
```

Q3: Student result

Welcome result.py

result.py

```python
import sys
mark1=float(sys.argv[1])
mark2=float(sys.argv[2])
mark3=float(sys.argv[3])
total=mark1+mark2+mark3
average=total/3
print("Mark1 =",mark1)
print("Mark2 =",mark2)
print("Mark3 =",mark3)
print("Total =",total)
print("average=",average)
if mark1 >= 40 and mark2 >= 40 and mark3 >= 40 :
    print("Result=Pass")
else:
    print("Result=Fail")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
[master (root-commit) 3f12d7c] student result
 1 file changed, 15 insertions(+)
 create mode 100644 result.py
PS C:\Users\batch1\Downloads\3exercise> git remote add origin https://github.com/Nirmal-Priya/3
PS C:\Users\batch1\Downloads\3exercise> git branch -M main
PS C:\Users\batch1\Downloads\3exercise> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 20 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 394 bytes | 394.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Nirmal-Priya/3question.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\batch1\Downloads\3exercise>
```

```
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 3f12d7c1d850d9ca5b8ab6615bdd2e2ccd948c12 # timeout=10
Commit message: "student result"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\3exercise>python result.py 80 90 75
Mark1 = 80.0
Mark2 = 90.0
Mark3 = 75.0
Total = 245.0
average= 81.66666666666667
Result=Pass
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Git branch

main

mark1

mark1 value

80

mark2

mark2 value

90

mark3

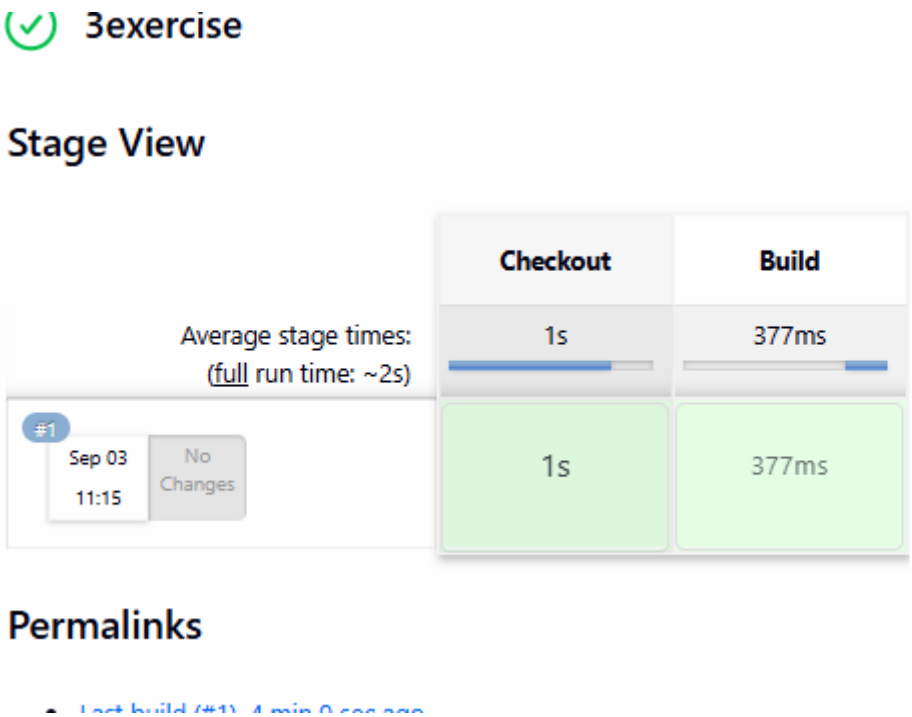
mark3 value

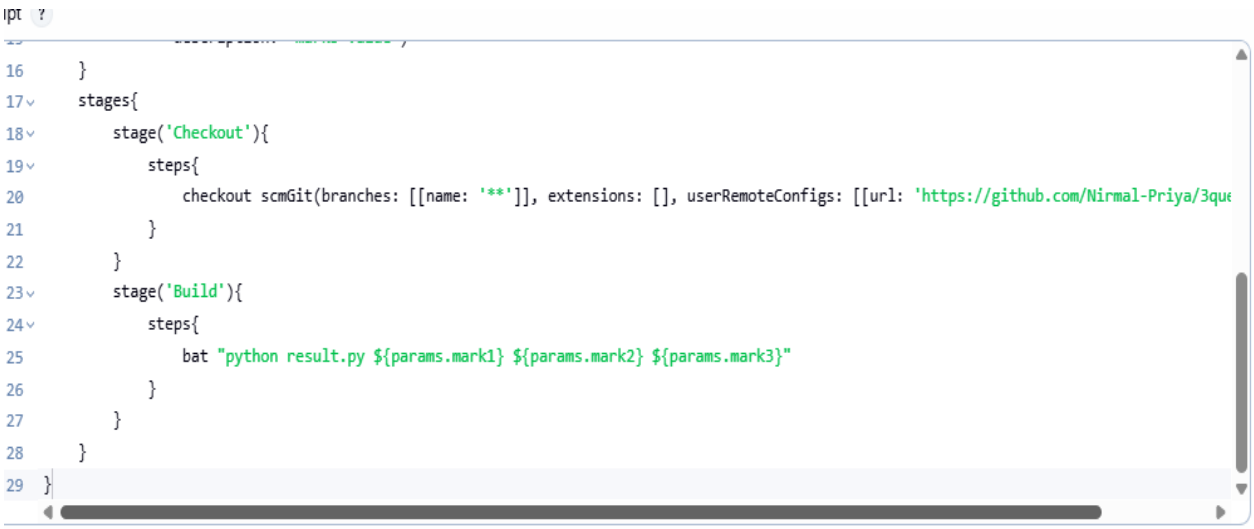
75

BRANCH NAME

Git Branch

main






```
pipeline{
   agent any
    parameters {
      string(name: 'BRANCH_NAME',
            defaultValue: 'main',
            description: 'Git branch')
      string(name: 'mark1',
            defaultValue: '80',
            description: 'mark1 value')
      string(name: 'mark2',
            defaultValue: '90',
            description: 'mark2 value')
```

```
        string(name: 'mark3',
               defaultValue: '75',
               description: 'mark3 value')
    }
    stages{
        stage('Checkout'){
            steps{
                checkout scmGit(branches: [[name: '**']], extensions: [],
userRemoteConfigs: [[url: 'https://github.com/Nirmal-Priya/3question.git']])
            }
        }
        stage('Build'){
            steps{
                bat "python result.py ${params.mark1} ${params.mark2} $
{params.mark3}"
            }
        }
    }
}
```

```
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 3f12d7c1d850d9ca5b8ab6615bdd2e2ccd948c12 # timeout=10
Commit message: "student result"
 > git.exe rev-list --no-walk 3f12d7c1d850d9ca5b8ab6615bdd2e2ccd948c12 # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\3exercise>python result.py 35 40 50
Mark1 = 35.0
Mark2 = 40.0
Mark3 = 50.0
Total = 125.0
average= 41.666666666666664
Result=Fail
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```